

UBCx Phot1x Final Report

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Abstract: Design, simulation, and preliminary experimental characterization of a Mach-Zehnder Interferometer (MZI) implemented in a 500 nm × 220 nm silicon strip waveguide platform is reported. The waveguide group index from spectral measurements is extracted and compared with values obtained from Lumerical MODE corner analysis simulations. A device with non-optimal routing was measured, yielding a group index of 4.285. The result lies slightly outside the original simulated corner analysis range (4.18 -- 4.26).

Introduction:

Silicon photonics has emerged as a key technology for high-speed optical communication and sensing applications. The Mach-Zehnder Interferometer (MZI) is one of the most fundamental building blocks in this platform, commonly used in modulators, filters, and sensors. Accurate knowledge of the waveguide group index is essential for predicting MZI performance, particularly the Free Spectral Range (FSR).

The following project aims to design an MZI from which the group index can be experimentally extracted and compared against waveguide simulations that account for fabrication variability through corner analysis. The design is based on a standard 500 × 220 nm strip waveguide operating in the quasi-TE polarization. The chip included various designs of the MZIs with varying waveguide pathlengths, but unfortunately only one fabricated device with incorrect routing between the Y-branch and the balanced directional coupler (BDC) was measured and analyzed.

Theory:

An unbalanced MZI consists of an input splitter, two waveguide arms of different lengths, and an output combiner. In this work, a Y-branch was used as the splitter and a BDC was used as the combiner. Light is coupled on and off the chip using grating couplers (GCs).

The transmission spectrum of a lossless MZI is governed by the interference between two optical paths with a length difference ΔL . The output intensity can be expressed as:

$$T(\lambda) = \frac{1}{2} [1 + \cos(\frac{2\pi n_{eff}(\lambda) \Delta L}{\lambda})] \quad (1)$$

The Free Spectral Range (FSR), defined as the wavelength spacing between adjacent transmission peaks, is given by:

$$FSS(\lambda) = \frac{\lambda^2}{n_g(\lambda) \cdot \Delta L} \quad (2)$$

Where n_g is the group index of the waveguide. Rearrangement of this equation allows extraction of the group index directly from measured spectral data:

$$n_g = \frac{\lambda^2}{FSR \cdot \Delta L} \quad (3)$$

This relation forms the basis for the experimental group index extraction performed in this course.

Waveguide Modeling and Simulation:

The general parameters for all variations of the waveguide were 500 x 220 nm (w x h) with quasi-TE polarization. The design layout of the waveguides is shown in Figure 1 below:

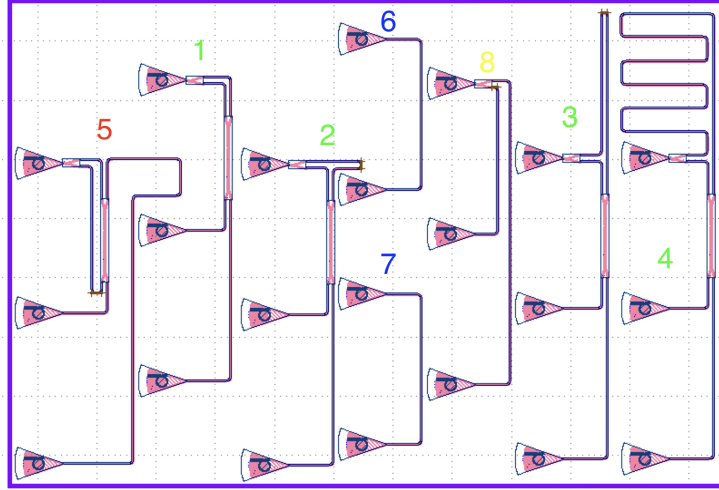


Figure 1: Design layout of the unbalanced 500x220nm MZIs containing four varying length MZI designs (labelled in green 1-4), one incorrectly routed device which was measured (5 in red), 2 demultiplexed devices (6,7 in blue)

The simulated mode profiles of such device was simulated using LUMERICAL MODE and is shown in Figure 2:

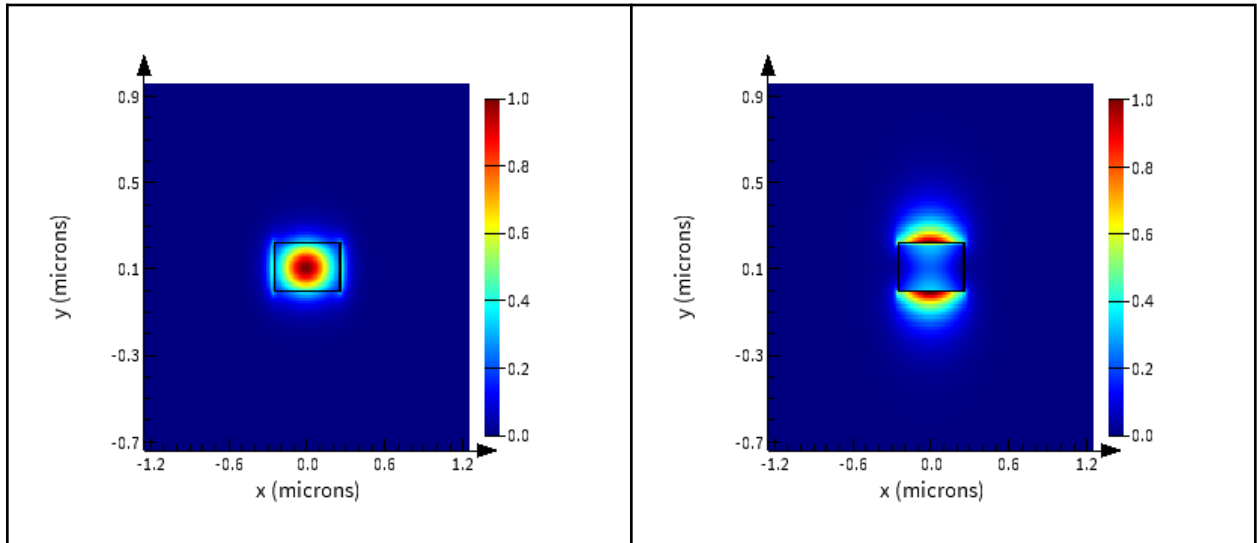


Figure 2: TE (left) and TM (right) mode profiles at 1.55 um

The effective index was fitted to a second-order Taylor expansion around $\lambda_0 = 1.55 \mu\text{m}$ using Lumerical MODE frequency sweep data:

$$n_{\text{eff}}(\lambda) = a_1 + a_2(\lambda - \lambda_0) + a_3(\lambda - \lambda_0)^2$$

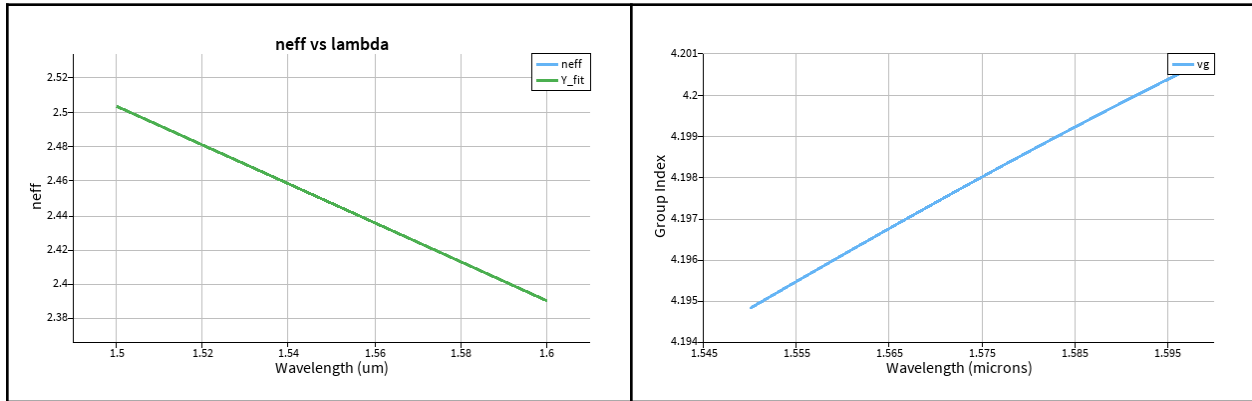


Figure 3:

Fabrication variability was accounted for by performing corner analysis simulations in Lumerical MODE with varying parameters; height in ranges of 215.3 – 223.1 nm and widths in ranges of 470 – 510 nm. As a result the group index for the Si waveguide in TE polarization was found to be in the range of 4.18 – 4.26 at 1.55 μm .

MZI Design and Simulation:

Four different MZIs were designed with varying path lengths as shown in Figure 1. Unfortunately, only device 5 (which was a nonsensical device i forgot to remove from the chip) was measured and obtained results for.

Table1: Mach-Zehnder Interferometers in the EBeam_ksenia257_new chip with their respective path lengths and FSR

Device	ΔL (um)	FSR (nm)
1 (opt_in_TE_1550_device_MZI1)	10.2	57.27
2 (opt_in_TE_1550_device_MZI2)	55.08	10.41
3 (opt_in_TE_1550_device_MZI3)	258.84	2.22
4 (opt_in_TE_1550_device_MZI4)	664.84	0.86
5 ((opt_in_TE_1550_device_MZI) - measured	80.66	~6.95
Ideal MZI	100	~ 5.7

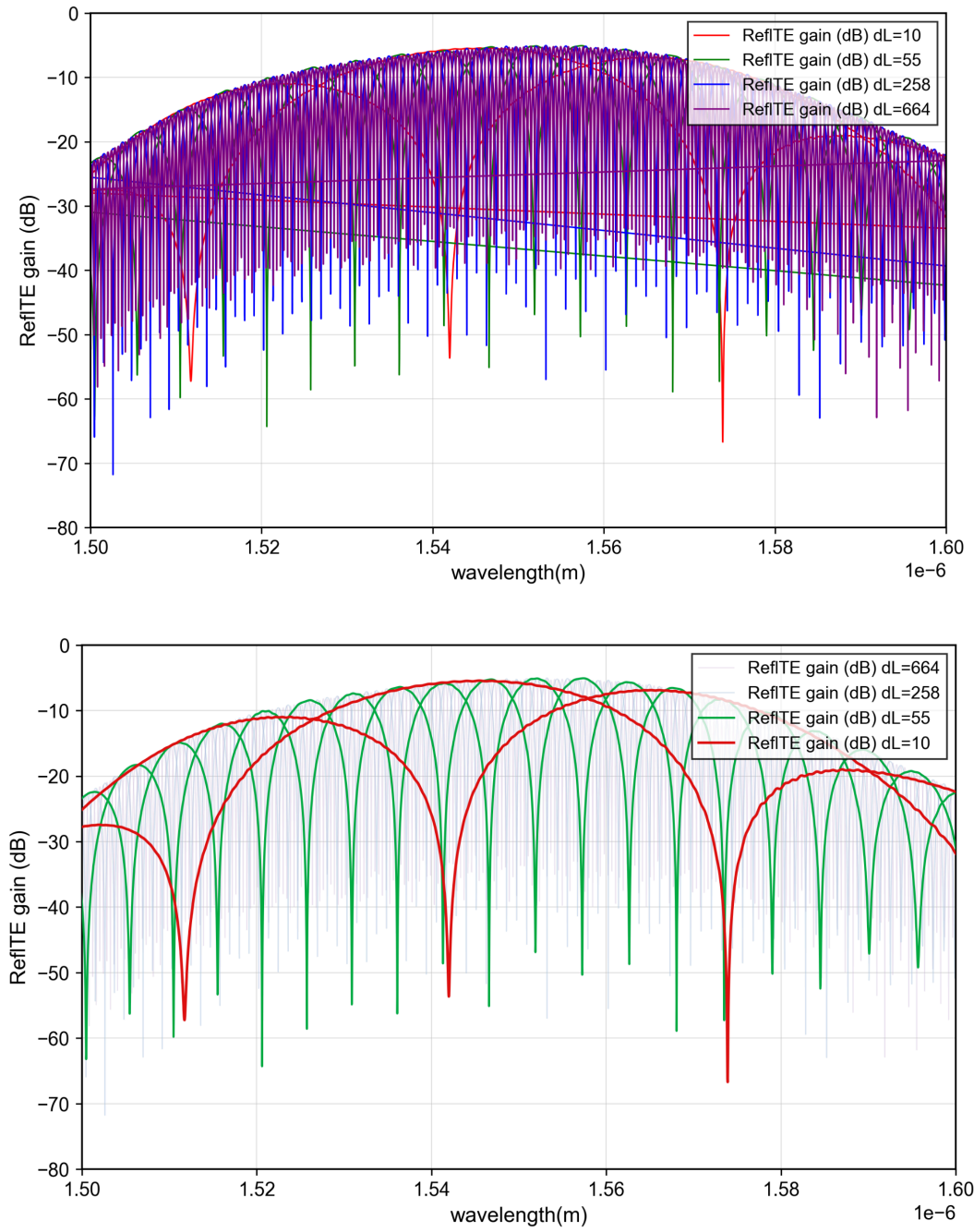


Figure 4: Transmission spectra (top and bottom) of the MZIs for four different path lengths and for the fundamental quasi-TE mode.

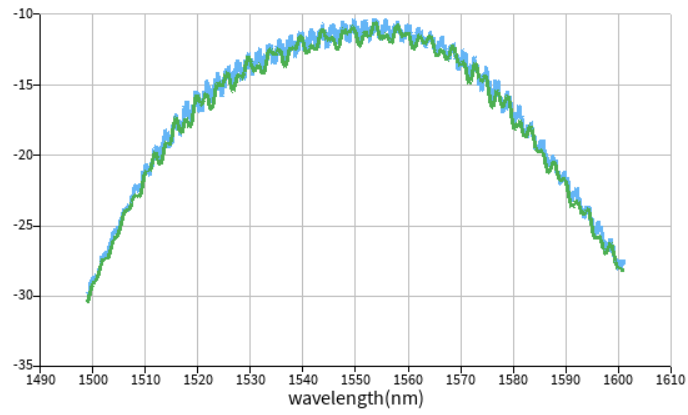


Figure 5: Transmission spectrum of the measured device 5

Experimental Results:

Experimental results were obtained at x, using y, etc only for device 5. The spectrum can be seen below:

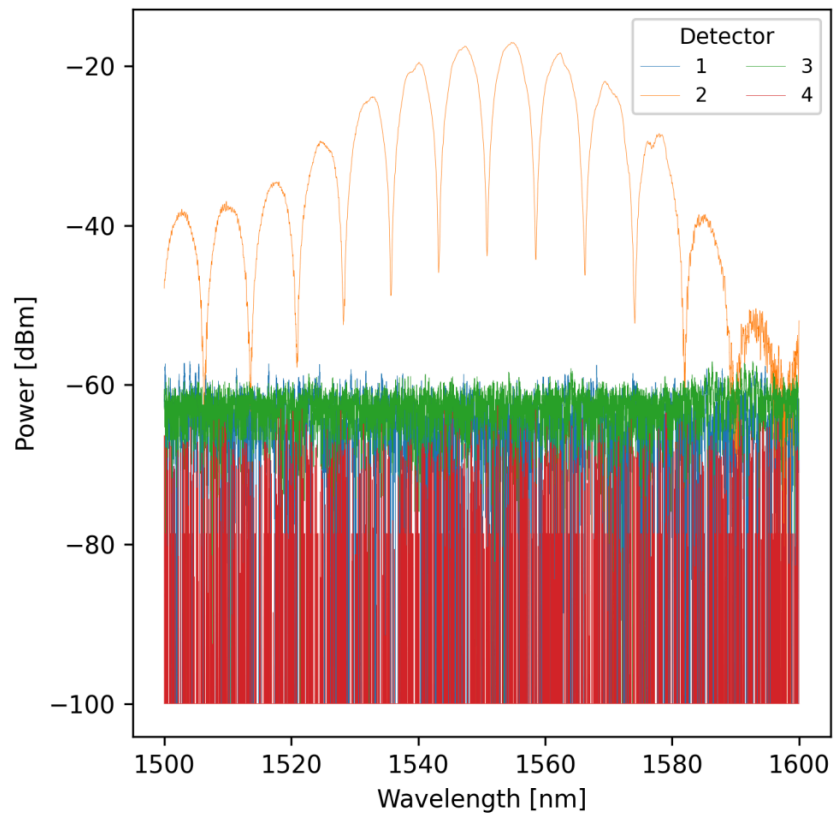


Figure 6: Experimental results for device 5

Curve fitting of the measured device yielded $n_g = 4.285$ at $1.55 \mu\text{m}$, free spectral range of $\approx 6.95 \text{ nm}$

Results Analysis and Discussion:

The group index extracted from the measured device was 4.285 at 1550 nm, with a corresponding Free Spectral Range of approximately 6.95 nm. This value lies slightly above the maximum group index obtained from the original corner analysis simulations (4.18 – 4.26). Several factors may contribute to this small discrepancy; the measured device (opt_in_TE_1550_device_ksenia257_MZI) featured non-optimal routing between the Y-branch and the balanced directional coupler. The waveguide arms were connected to the top and bottom ports of the BDC rather than the conventional left-side ports. This layout error likely introduced additional phase errors and reduced fringe visibility, which in turn affected the accuracy of the curve fitting process. Another consideration is fabrication variations may have resulted in waveguide dimensions that lie slightly outside the ranges initially assumed in the corner analysis. To reconcile the experimental result with simulation, an expansion of the fabrication variation parameters can be considered; waveguide width from 460–520 nm and height from 214–225 nm. With this adjustment, the maximum simulated group index increases to ~ 4.29–4.30, which consequently encompasses the measured value. It should be noted that de-embedded loopback structures were included in the chip design. However, de-embedding of the grating coupler insertion loss was not performed prior to curve fitting. This was primarily because only the non-ideal device was measured, and its routing limitations already constrained the quality of the spectral data. Proper de-embedding would be beneficial for future measurements of the other MZI structures on the chip (MZI1–MZI4), which feature conventional routing. Overall, while the non-ideal layout of the measured device limited the precision of the extracted parameters, the results still demonstrate that group index extraction from MZI spectra is feasible even under these conditions. The extracted value of 4.285 remains close to the simulated range and provides useful insight into the fabricated waveguide properties :)

Conclusions:

A Mach-Zehnder Interferometer was designed and fabricated in a 500×220 nm silicon strip waveguide platform. Waveguide compact models and corner analysis were developed using Lumerical MODE. Experimental characterization of the only measured device yielded a group index of 4.285, which lies slightly outside the original simulated range. A modest expansion of the assumed fabrication variation is proposed to reconcile simulation and experiment. Despite the non-ideal layout of the measured device, the results demonstrate the feasibility of extracting waveguide parameters from MZI spectra.

References:

[1] L. Chrostowski and M. Hochberg, *Silicon Photonics Design: From Devices to Systems*, Cambridge University Press, 2015.

[2] Course materials – UBCx Silicon Photonics Design, Fabrication and Data Analysis.

Acknowledgements:

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